



Orodromeus laid up to 24 eggs.

The first dinosaurs

One of the earliest dinosaurs was *Eoraptor*, meaning "early plunderer". It was no bigger than a large dog and lived 225 million years ago (mya). As with all the early dinosaurs, it was a carnivore and walked on two legs.



Orodromeus nest

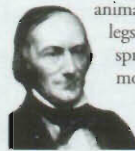
Eoraptor skull

Breeding

Dinosaurs laid hard-shelled eggs as some reptiles do today. Many dinosaurs laid a clutch of eggs in a hollowed-out nest in the ground. Several fossilized nests have been found close together, which suggests that some dinosaurs nested in colonies. The chicks developed rapidly and may have left the nest soon after hatching. Many were cared for by the parent dinosaur until they were able to look after themselves.

Richard Owen

Born in Lancaster, England, Richard Owen (1804–92) became the Hunterian Professor of the Royal College of Surgeons in 1836. As well as being an anatomist, he was a brilliant palaeontologist. He was the first to use the term "dinosaurs", which means "terrible lizards", in a report in 1842. He noted that these animals had pillar-like legs, rather than the sprawling legs of modern reptiles, and should be classified separately.



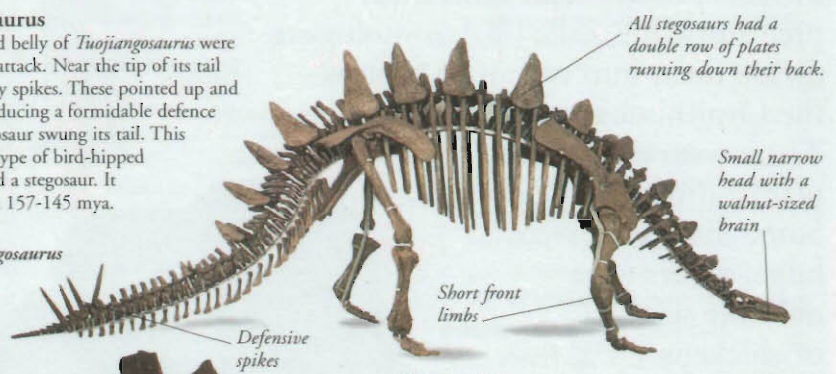
Defence

Dinosaurs protected themselves against attack from predators. Different dinosaurs developed a variety of powerful defences. For example, *Triceratops* had horns on its head, *Euoplocephalus* had a tail club, and *Tuojiangosaurus* had a spiky tail. Some of these adaptations may have had several functions, but one of them was likely to have been defence. Scientists cannot say exactly how these animals defended themselves, but it is easy to imagine.

Tuojiangosaurus

The flanks and belly of *Tuojiangosaurus* were vulnerable to attack. Near the tip of its tail were four bony spikes. These pointed up and outwards, producing a formidable defence when the dinosaur swung its tail. This animal was a type of bird-hipped dinosaur called a stegosaur. It lived in China 157–145 mya.

Tuojiangosaurus



All stegosaurs had a double row of plates running down their back.

Small narrow head with a walnut-sized brain

Short front limbs

Defensive spikes

Thumb spike

Raised nodules for protection

Iguanodon spike

When *Iguanodon* was first reconstructed, its large spike was placed on its beak. It is now known that the spike was on its thumb and may have been used as a defensive weapon against predators. The spike could have pierced the belly, throat, or eye of an attacker. The dinosaur may also have used it in fights for status with other *Iguanodons*, and even to help it feed.

Dinosaur skin

Occasionally, the skin, or skin impression, of dinosaurs is preserved. From these fossils we can tell that the skin of many dinosaurs was not smooth, but nodular and rough. This would have given some protection against the claws and teeth of predators. This is the skin of *Polacanthus*.

Euoplocephalus had thick bone plates and spikes over its back, with a large shoulder spike for added protection.

Reconstruction of Iguanodon hand



Claw

Brow horn

Frill anchored the jaw muscles.

Euoplocephalus

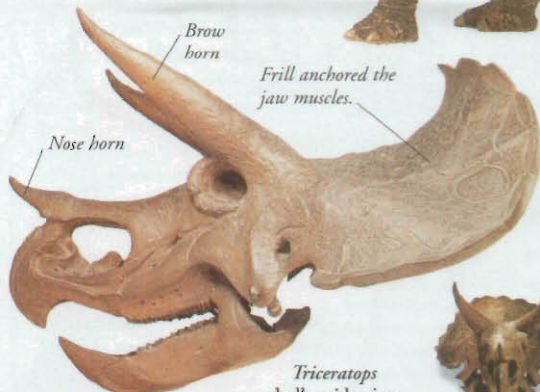
Club was made out of several bones fused together.

Euoplocephalus

This armoured ornithischian had a large bony club at the tip of its muscular tail. It could have swung this with great force, disabling a predator.

Triceratops

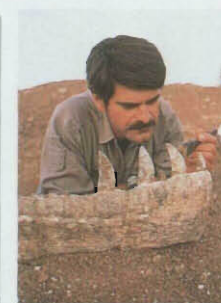
The ceratopsians, or horned dinosaurs, were ornithischians. Most of them had brow horns and nose horns. *Triceratops*, the largest ceratopsian, had two long horns on its brow, a short nose horn, and also a bony neck frill protecting its neck. Its head took up nearly one-third of its length. It probably used its horns to fend off predators, and males used them to deter rivals in the herd, mostly by display, but also by fighting.



Triceratops skull – side view



Triceratops skull – front view



Dinosaur discoveries

Removing dinosaur fossils from surrounding rock is tricky. Some need to be protected in a jacket made of plaster or polyurethane foam before they are taken to a laboratory. Fossils are found every year, and each discovery teaches us more about these extinct animals.

Finding dinosaur bones.

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